



Crop observations

4 wheat sites · 5 maize sites
Growth parameters and site-year partitions



Wuqiao irrigation experiment

Water calibration: 2016–2018
Retrospective testing: 2019

Continuous regional rotations

Shared crop parameters · weather · soil · rotation area
32 representatives × 5 irrigation levels



Spatial irrigation allocation

Uniform versus targeted
Identical regional irrigation budgets
Selection: 1997–2013



Annual irrigation strategies

Storage-rainfall versus rainfall-only
Matched monitoring availability
Selection: 2003–2013

Comparison: 2014–2025

Rotation grain · irrigation · ET · drainage · spatial gains and losses